

15. Informal covariation assessment: Data-based versus theory-based judgments

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The flow of social experience frequently challenges us to recognize empirical covariations. Sometimes, these covariations are merely another test of our powers of observation and are of no immediate practical concern to us. At other times – for example, when those covariations involve early symptoms of problems and later manifestations, or behavioral strategies employed and outcomes obtained, or relatively overt characteristics of people or situations and relatively covert ones – such detection abilities may help to determine our success in adapting to the demands of everyday social life. More generally, covariation detection will play a large role in our continuing struggle as “intuitive scientists” (see Nisbett & Ross, 1980; Ross, 1977, 1978) to evaluate and update the hypotheses we hold about ourselves, our peers, and our society. An obvious question therefore presents itself: How proficient are we, as laypeople, at assessing the empirical covariations presented by experiential evidence?

Before proceeding to discuss past or present research, we should note that everyday observation provides a great deal of relevant evidence; and it hints that the answer to the proficiency question is apt to be far from a simple one. On the one hand, both the generally adaptive nature of social behavior and the generally harmonious quality of social interaction leave little doubt that the participants in our culture possess many profound

We wish to thank Professors Daniel Kahneman, Amos Tversky, and Paul Slovic for their helpful comments and suggestions. We also wish to acknowledge the stalwart efforts of Rita French and Julia Steinmetz who were co-investigators for some of the research reported herein. The research was supported in part by a National Science Foundation Graduate Fellowship to the first author and by National Institute of Mental Health Grant MH-26736 and National Science Foundation Grant BNS-78-01211 to Lee Ross and Mark Lepper.

The organization and content of this chapter owe an obvious debt to an earlier treatment of the same topic by L. Ross and R. E. Nisbett in *Human Inference: Strategies and Shortcomings of Social Judgment*, Prentice-Hall, Inc., 1980.

insights about behavioral causes and consequences. In fact, contemporary attribution theorists (e.g., Jones et al., 1971; Kelley, 1967, 1971, 1973; Weiner, 1974), who have been uniquely concerned with the basis for such insights, generally seem to have treated the ability to detect covariation as a “given” and proceeded to the inferential use that is made of such perceived covariations. On the other hand, everyday experience also offers eloquent testimony to the existence of ill-founded prejudices, ruinous behavioral strategies, and other manifestations of erroneous causal or correlational theories. It should be apparent, therefore, that any adequate portrait of informal covariation assessment must address the unevenness of the layperson’s performance. It must somehow reconcile subtle insights about some functional relationships with ignorance or illusions about other relationships.

Prior investigation of covariation assessment

It is noteworthy, perhaps, that even the staunchest defenders of the layperson’s capacities as an intuitive scientist (e.g., Peterson & Beach, 1967) have had little that was flattering to say about the layperson’s handling of bivariate observations. We shall review two lines of research that support this unflattering portrait, focusing in each case on procedural aspects and results that will prove pertinent to the subsequent report of our own research on lay covariation assessment.

Reading contingency tables: The “ecological validity” issue

One important line of prior research has dealt with people’s ability to recognize functional relationships presented in simple 2×2 contingency tables. Typically, these tables summarized the number of instances of the presence and absence of variable *X* (for example, a particular disease) purportedly associated with the presence and absence of variable *Y* (for example, a particular symptom). With bivariate data so neatly assembled and “packaged,” the subjects’ task seemingly was an unusually simple and straightforward one. Nevertheless, the evidence (e.g., Jenkins & Ward, 1965; Smedslund, 1963; Ward & Jenkins, 1965), shows that laypeople generally have fared quite poorly at such tasks.

Judgmental strategies that ignore one or more of the four cells proved to be at the root of most subjects’ difficulties. One common failing, for instance, involves the virtually exclusive reliance upon the size of the “present-present” cell relative to the entire population. Thus many subjects may say that symptom *X* is associated with disease *A* simply because the contingency table reports a relatively large number of cases in which people do, in fact, have both the disease and the symptom. Subjects paying attention to only two of the four cells may similarly be misled. Noting, for instance, that more people with the disease lack symptom *A*

than possess it, they may therefore conclude that the relevant relationship is a negative one; or noting that more people with the symptom have the disease than are free of it, they may conclude that the relationship is a positive one.

Without formal statistical training, however, very few people appreciate the fact that valid inferences in such cases can be made only by considering *all* of the four cells. One satisfactory method, for example, might involve comparing *proportions* (i.e., comparing the proportion of diseased people manifesting the particular symptom with the proportion of non-diseased people manifesting that symptom). The formal strategy dictated by contemporary texts on statistical inference, of course, would similarly involve attending to all four cells. Specifically, one would be required to note discrepancies between observed cell frequencies and those frequencies to be “expected” on the basis of the relevant marginal frequencies or probabilities (thereby providing the information required for calculation of a chi-square statistic).

Skeptics inclined to champion the layperson’s cause might be tempted to dismiss such demonstrations as evidence only that people are poor at “reading” contingency tables and to contend that the subjects’ errors in such tasks primarily are products of the task’s novelty and artificiality. The relationship between laboratory performances and everyday real-world performance – that is, the issue of “ecological validity” – is an important problem and one that cannot be evaded in the research reported in this chapter. Let us, accordingly, begin to address the issue here in the hopes that the reader will be forewarned and forearmed.

First, while the table-reading task is undeniably unusual and ecologically unrepresentative, let us note that the logic (and the various logical shortcomings) displayed by subjects in those laboratory tasks is quite consistent with that displayed in a variety of everyday inferences. Consider, for example, the perplexing question, “Does God answer prayers?” “Yes,” the layperson who consults only the present-present cell may answer, “because many times I’ve asked God for something, and He’s given it to me.” His more sophisticated and skeptical peer is apt to ask, “But how often have you asked God for something and *not* gotten it?” Comparison of two cells, however, still is utterly inadequate for the inferential task at hand. Even the addition of a third cell – favorable outcomes that occurred in the absence of prayer – would leave the issue in doubt, for all four cells are required before any solid inference can be made. And even the most sophisticated of intuitive psychologists would probably balk at the suggestion that data from the “absent-absent” cell (i.e., favorable outcomes that were not prayed for and that did not occur) are indispensable for assessing the impact of prayers on worldly outcomes.

The charge of non-representativeness can be answered even more forcefully, however. If the fourfold-table task creates some unique prob-

lems the same task also *sparcs* subjects most of the demands that contribute to the difficulty of everyday covariation assessments. These demands include initially sampling, coding, storing, and retrieving the relevant data and then arranging them in a form that permits covariation assessment. Consider, for instance, the man who undertakes to test the stereotyped notion that red-haired people are hot-tempered (or, more precisely, that the presence of red hair is positively correlated with the presence of a hot temper). First, he must decide which data are to be considered. Shall he attempt to sample or recall instances of hot-headed red-haired people? Or shall he consider some red-haired people and note how many are hot-tempered, or some hot-tempered people and note how many have red hair? Very likely he will adopt some such strategy or combination of strategies without ever entertaining the possibility that the relative frequency of even-tempered brunettes might be relevant to the inferential task at hand. This state of affairs contrasts with the fourfold-table task that makes all of the relevant data equally available to the subject, even if it does not demand that they all be given proper subsequent consideration.

Let us suppose that the individual in question somehow does recognize that testing his stereotype demands that he consider the data in all four cells. How shall he draw an appropriate data sample from which to generate the relevant cell frequencies? Should he simply consider the first set of people who happen to come to mind? Should he consider all of the people in some restricted class (for instance, his family members, his circle of friends, or the tenants in his apartment building)? Or should he attempt to adopt some "random" method for generating cases from the overall population? The potential sources of bias in most informal sampling methods are obvious. Samples of the people who first come to mind are apt to show serious "availability" biases (Tversky & Kahneman, 1973, 11). And samples of friends or family members or neighbors are likely to show the same bias, as well as violating the independence requirement (not a trivial problem, since one's family, one's friends, or even one's neighbors may tend to share behavioral tendencies, physical characteristics, and even correspondences between them). The "novel" fourfold-table procedure, of course, spares the covariation assessor the task of confronting these perplexing and potentially crippling problems.

The tasks of data coding, storage, and retrieval bring a host of additional factors and problems into play in everyday covariation assessment. Most importantly, the implicit hypothesis that redheads are hot-tempered may bias decisions about who is or is not hot-tempered and whose hair is or is not red. This issue receives further attention later in this chapter. For now let us merely note that one's prior expectations or hypotheses constitute a potentially biasing influence on every stage of information processing, an influence that is precluded when one is presented with data for which all of the processing involved in coding, storing, and retrieving have already been completed. More generally, let us simply reiterate that such real-

world impediments to accurate covariation assessment are apt to prove far more forbidding than any arising from the artificiality or unfamiliarity of the contingency-table methodology.

Illusory correlation: The impact of preconceptions on perceptions

How successful are people at dealing with bivariate distributions when the relevant observations are presented individually rather than prepackaged in the form of contingency tables? Much of the existing evidence derives from the Chapmans' seminal work on illusory correlation. For a detailed account of this research see Chapman & Chapman, 1967, 1969; also see chapter 17, this volume. Our brief review here focuses on the particular issues and findings that set the stage for our own, more recent, efforts.

Perhaps the simplest summary of the Chapmans' findings and conclusions about subjects' ability to detect covariations between clinical signs and symptoms is that reported covariations seem to reflect true covariations far less than theory-based or semantically based preconceptions about the nature of the relationships that "ought" to exist. While these findings were dramatic, controversial, and of considerable immediate relevance to the practitioner, the general point they made about the effects of preconceptions upon observed associations was essentially a familiar one.

As L.J. Chapman himself noted in his initial (1967) paper, illusory correlations are not restricted to the domain of clinical judgment. Most superstitions essentially are empirically groundless beliefs about the association between particular actions or events and subsequent positive or negative outcomes. Racial, ethnic, regional, religious, or occupational stereotypes similarly are beliefs about covariations, beliefs that are strongly held and remarkably resistant to the impact of non-supporting data (cf. Adorno et al., 1950; Allport, 1954; P. Goldberg, 1968; J.M. Jones, 1972; Taynor & Deaux, 1973). Research on "implicit personality theory" can also be cited in this context. Over half a century ago Thorndike described the so-called halo effect, whereby perceivers expect and report the association of all positive personality characteristics. Subsequent researchers (e.g., Koltuv, 1962; Norman & Goldberg, 1966; Passini & Norman, 1966) have greatly extended this work and explored the conceptual schemas and beliefs about personality that dispose perceivers to report consistent relationships among different traits or different indicators of the same trait whose empirical linkages are very weak or even non-existent.

There is thus a wealth of empirical and anecdotal evidence that, in the realm of covariation assessment, the contest between expectations and evidence is apt to be an unequal one. Just as in other types of perceptual and cognitive judgments (cf. Bruner, 1957a, 1957b; Bruner, Postman, & Rodrigues, 1951), the intuitive scientist's preconceptions about empirical rela-

tionships are apt to determine what he detects, what he fails to detect, and what he sees that is not really there to be seen. Such facile generalizations about the supremacy of theory over data, however, can lead us to overlook an important ambiguity – or at least an important unresolved question – in the Chapmans' studies. Specifically, we can identify two separate issues of interest about intuitive covariation assessment that are, in a sense, confounded in those studies and in subsequent follow-ups (e.g., Golding & Rorer, 1972; Starr & Katkin, 1969). The first issue involves people's difficulties in detecting covariations presented in immediately available data – that is, their capacity to recognize and assess covariations when they are "unencumbered" by any specific theories or expectations about the empirical relationship in question. The second issue involves the tendency for the subjects' intuitions or theories to lead them to assume the existence of strong correlations where such assumptions are belied by the best "objective" evidence.

The separation of these issues suggests the need to investigate subjects' covariations assessment performances in two very different tasks. The first task is essentially a "psychophysical" one yielding evidence about the relationship between subjective judgments and objective measurements. In this task subjects are simply to be presented with sets of bivariate observations about which they can hold no preconceived theories and then to be asked to assess the strength of relationship in each set. The investigator can then attempt to measure the difficulty of the subjects' task (primarily by noting the degree of variability associated with particular judgments) and to determine the nature of the "psychometric" function relating the subjective estimates to the objective measurements.

The second task deals with the opposite extreme to such purely "data-based" assessments; it concerns assessments that are exclusively, or almost exclusively, "theory-based." Thus pairs of variables or measurements that can be linked by the subjects' intuitions or theories are to be specified, but no relevant bivariate data are to be furnished by the investigator. If any data are brought to bear by subjects in such assessments, therefore, they must be supplied from the subjects' everyday experiences and recollections; and as such they are susceptible to the host of "processing" errors and biases alluded to earlier in our discussion. For purposes of evaluation and comparison, the experimenter also must have some means of assessing the accuracy, or the nature of the errors, associated with such intuitive assessments. That is, he must possess some "objective" measure of correlation for each pair of variables specified to the subjects.

Experimental comparison of data-based and theory-based covariation assessment

Recently we conducted a study pursuing the issues and employing the basic strategy just outlined. For the data-based task, three different types of

bivariate distributions, described in more detail in the next section, were employed. For the theory-based assessments, pairs of associated variables were specified but no actual data were presented by the experimenter. To facilitate comparisons between data-based and theory-based assessment, a within-subjects design was used in which subjects employed the same subjective rating scale for both tasks (with counterbalancing for order of presentation).

In each case, continuous variables rather than dichotomous ones (in contrast to the work of Chapman & Chapman, 1967, 1969, and Jenkins & Ward, 1965) were used in the relevant covariation assessment tasks. This feature was designed to capture more faithfully the nature of most everyday assessment tasks.¹

Subjects and rating tasks

A total of 64 Stanford undergraduates, none of whom had completed a college-level course in statistics, participated in this experiment designed to compare data-based and theory-based assessments of covariation. For both types of assessments, subjects used a simple rating scale to describe their subjective impression of the direction and strength of relationships between pairs of variables. In using this scale, subjects first estimated whether the relationship in question was positive or negative. They then estimated how strong the relationship was by placing an "X" on a 100-point rating scale with end points anchored with "perfect relationship" and "no relationship."

Data-based covariation estimates. One part of the experiment dealt with covariation estimates based on sets of bivariate observations provided by the experimenter. Three sets of stimulus materials were employed to create the relevant bivariate distributions. For each type of stimulus material, a set of bivariate distributions was created to represent different objective covariations between two variables. Subjects were asked to study each distribution and estimate the relationship it portrayed using the 100-point subjective rating scale.

We anticipated that the accuracy of subjects' estimates in this task would depend in part on the difficulties they had in coding, processing, and remembering the data. To test this assumption, the three types of stimulus materials differed markedly in the information-processing demands they placed upon subjects prior to covariation estimation, particularly with respect to the demand for estimation of stimulus magnitudes and for

¹ Our literature search revealed that covariation estimation tasks featuring continuous variables have very rarely been employed by previous investigators. The exception uncovered in that search was one rarely cited study (reported by Erlick, 1966; Erlick & Mills, 1967) that dealt primarily with the effects of variance on perceived relatedness and yielded very little data relevant to present concerns.

Table 1. Pairs of variables described to raters and their objective correlations

	Variables specified	Objective correlation
A	Students' self-ratings of liberalism Students' ratings of the performance of business leaders in the past decade	-.28 ^a
B	Students' self-ratings of intellectualism Students' ratings of the performance of U.S. presidents in the past decade	-.19 ^a
C	Students' self-ratings of shyness Number of U.S. states visited by students	-.12 ^a
D	Students' self-ratings of ambitiousness Students' heights	.01 ^a
E	Students' ratings of the performance of university presidents in the past decade Time spent per week by students on recreational athletics	.08 ^a
F	Students' self-ratings of intellectualism Students' family incomes	.17 ^a
G	Children's dishonesty as measured by false report of athletic performance Children's dishonesty as measured by amount of cheating in solving a puzzle	.18 ^b
H	Students' family incomes Students' ratings of the performance of business leaders in the past decade	.28 ^a
I	Sixth graders' ability to "delay gratification" Sixth graders' ability to resist temptation to cheat	.31 ^c
J	Students' self-ratings of conscientiousness Ratings of those students' conscientiousness by roommate	.35 ^d
K	Students' self-ratings of intellectualism Students' self-ratings of ambitiousness	.37 ^a
L	Students' ratings of Congress's performance in the past decade Students' ratings of labor leaders' performance in the past decade	.40 ^a
M	Time spent per week by students on recreational athletics Students' self-ratings of physical fitness	.52 ^a
N	Students' ratings of U.S. presidents' performance in the past decade Students' ratings of business leaders' performance in the past decade	.55 ^a
O	Self-ratings of political conservatism A composite of self-rating items from the Ethnocentricity scale	.57 ^c
P	Students' height Students' weight	.79 ^a

^aCorrelation obtained from a survey ($N = 295$) of Stanford University students.

^bCorrelation obtained from Hartshorne & May (1928).

^cCorrelation obtained from Mischel & Gilligan (1964).

^dCorrelation obtained from Bem & Allen (1974).

^eCorrelation obtained from Adorno et al. (1950).

storing and retrieving information from memory. Thus, the first type presented sets of 10 simple number pairs. The second type presented drawings, each portraying a set of 10 men of varying heights holding walking sticks of varying heights. The final type of distribution was presented in the form of audiotapes on which a set of 10 individuals were each heard first to say some letter occupying a particular ordinal position in the alphabet (purportedly the initial letter of their surname) and then immediately to sing a musical note of varying duration.

Theory-based covariation estimates. The other part of the experiment (order of presentation of the two parts was counterbalanced) dealt with covariation estimates based on subjects' a priori expectations or theories rather than any immediately available bivariate data.

Pairs of measurements or variables were specified, and subjects were required to estimate first the direction and then the strength of the relationship for each pair, always employing the same simple 100-point subjective rating scale as that employed for the data-based assessments. Some of these pairs in this part of the experiment dealt with divergent behavioral measures of personal dispositions, for example, two measures of honesty used in Hartshorne and May's (1928) classic study of cross-situational consistency in moral behavior; others dealt with personal attitudes, habits, or preferences. In each case, however, the subjective covariation estimates we obtained could be compared with "objective" correlations culled from previous empirical studies. (A complete list of these variables pairs and pertinent correlations is presented in Table 1.)

Results

Data-based covariation estimates

Our first concern involves the estimates of covariation that subjects made in response to immediately available bivariate data. Two specific questions can be addressed: First, how readily could individuals detect covariations of various magnitude in the absence of any "theory" about the relevant data distributions? Second, what was the nature of the "psychophysical function" relating the mean of the raters' subjective estimates to an objective measure of covariation?

Examination of the data suggested that the estimation task was very difficult. The standard deviations and interquartile ranges associated with the overall group estimates for each data set were extremely high (see Figure 1). What is particularly notable is the difficulty many subjects experienced in simply recognizing the existence of positive relationships, even those of respectable magnitude. In fact it is only when objective correlations reach the level of $+ .6$ to $+ .7$ that the interquartile interval consistently excludes negative estimates of the relationship. The instabil-

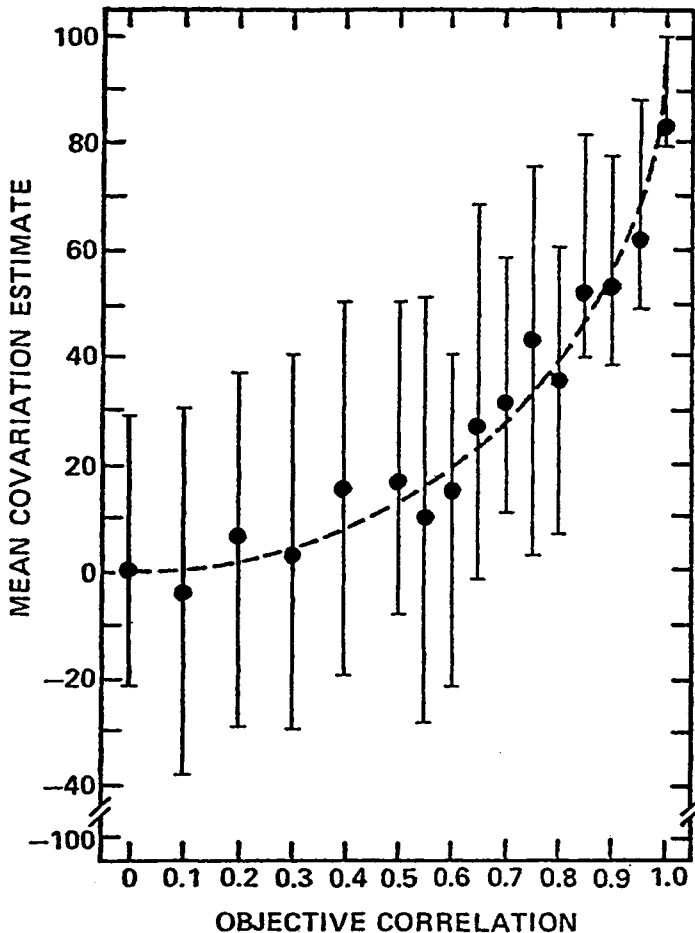


Figure 1. Means and interquartile ranges for "data-based" covariation estimates, pooling results for the three types of distribution. The "psychophysical function," $100(1 - \sqrt{1 - r^2})$, is shown as a dashed line.

ity of covariation estimates, incidentally, did not seem to differ systematically for the three types of bivariate data distributions, despite the apparent differences in the amount of information processing (and hence the possibilities for random or systematic error) associated with these three types. This apparent insensitivity to increased information-processing demands may offer some clues about the nature of the subjects' assessment "strategy" – a possibility that we pursue somewhat later in this chapter.

Also presented in Figure 1 are data relevant to the second question about data-based assessment, the question of the "functional relationship" between objective and subjective indicators of covariation. (The reader

will note that means for the three stimulus domains are pooled in this presentation of results. Neither inspection nor significance tests revealed any significant or consistent trends distinguishing the results for the three stimulus domains, and the result of such pooling is a smoother and more comprehensible function than that obtained for any of the three individual domains.) For the reader's convenience we have fit a curve to the points in our graph. This curve, we should emphasize, was drawn *post hoc*; it was not derived from any "psychophysical theory."

It is clear by inspection that, despite the within-group variability for individual estimates, the function relating mean subjective estimates to objective correlations (Pearson r 's) was a rather smooth one. More specifically, there seems to be a sharply accelerating function relating the two variables. Thus, relationships in the range commonly dealt with by psychologists concerned with measuring personality traits or other cross-situational consistencies in social behavior (i.e., $r = .2$ to $.4$) are barely detectable, yielding mean estimates in the range of 4 to 8 on the 100-point scale. Even relationships considered very strong by such psychologists (i.e., $r = .6$ to $.8$) result in rather modest subjective estimates of covariation. Objective correlations of $.7$, for instance, produced a mean subjective estimate of 34 – a rating midway between the points labeled "rather weak" and "moderate" on the 100-point subjective scale. Only when the objective correlations approached the $.85$ level did the group mean reach the midpoint of the subjective scale, and only beyond that point did subjects consistently rate the relationships as strongly positive.

A closer look at the estimates suggests that subjective ratings of relatedness or covariation are not a linear function of r or even r squared. Rather, the pattern of the pooled mean estimates seems well-captured by the expression, $1 - \sqrt{1-r^2}$. (The reader may recognize this expression as the "coefficient of alienation," a measure of the reduction in the standard error associated with prediction of variable Y based on knowledge both of variable X and the correlation between X and Y [Huntsberger, 1967].) Indeed, the similarity between the actual mean estimates and those described by this expression was quite striking – to be precise, 95% of the between-means variation is accounted for by the function based on the coefficient of alienation. We hasten to remind our readers, however, that they should not misinterpret the meaning of this neat fit. The goodness of fit portrayed in Figure 1 applies only to group estimates; individual estimates attested to the subjects' difficulties in distinguishing different levels of covariation.

It obviously is premature to conclude anything about the precise nature of the "psychophysical" function relating subjective responses to objective measures of covariation. Details of the task, context, and perhaps even the population of raters would undoubtedly produce differences in the function, differences that could be captured only through the inclusion of appropriate parameters. Nevertheless, the present results do probably

suffice to suggest that modest correlations of the sort so often reported by the cadre of personality assessors (cf. Mischel, 1968, 1969) are likely to go largely undetected or to make only the weakest of impressions on the layperson who encounters such covariations in the absence of theory-based preconceptions.

It is further worth reemphasizing that the three different data-based covariation estimation tasks produced remarkably similar results despite the greatly differing information-processing demands they placed upon subjects. Thus, the psychophysical function, illustrated in Figure 1, accurately reflects the relationship between subjective estimates and objective measures for *each* of the three tasks undertaken by subjects. The correlations between the means of subjects' actual estimates and the estimates predicted by the function range from $r = .91$ for the tone/letter pairs to $r = .98$ for the number pairs.

Theory-based estimates

Thus far we have presented evidence that relatively strong objective correlations (in terms of Pearson's r) are required to prompt subjective assessments even modestly different from zero. Considered in isolation, the psychophysical function for data-based assessments could simply be evidence that the layperson's subjective metric is a cautious or conservative one relative to that employed by most formal statisticians. (Certainly few would dispute that r^2 is a more appropriate index of relatedness than r , since it can so much more readily be related to predictive utility or reduction in uncertainty about the value of one variable based on knowledge of the other, associated, variable.) In short, the obtained data-based estimates do not suggest any systematic shortcoming on the part of the intuitive psychologist, although the variability associated with such estimates leaves little doubt about the difficulties and uncertainties of the estimation task.

With these findings and possible interpretations in mind, we can now turn to the results for subjective covariation estimates about the relationships listed in Table 1. These estimates, we should recall, were made in the absence of immediately available data; presumably, they were made only on the basis of the raters' informal theories or intuitions, with no data save that highly "processed" data noted and recalled from everyday experience. Figure 2 provides a first view of these results. It is clear, immediately, that no single function – linear or otherwise – captures the relationship between subjective estimates and objective measures. Nevertheless, there is an undeniable tendency for subjects' theory-based estimates to show a rough correspondence with the objective evidence. That is, positive empirical relationships were estimated to be positive; negative relationships were estimated to be negative; relatively strong empirical relation-

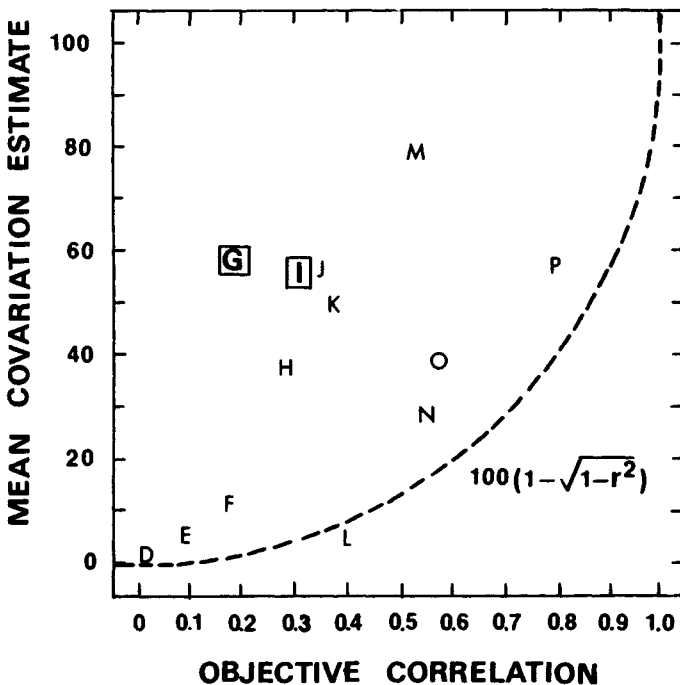


Figure 2. Means for "theory-based" covariation estimates. (See Table 1 for key to specified variables.)

ships generally were estimated to be stronger than relatively weak ones; and so forth.

Perhaps the most striking feature of these theory-based estimates, however, is that once freed from the constraints imposed by immediately available data subjects ceased to be cautious or conservative. They willingly ventured into the middle and even upper regions of the subjective 100-point scale even when dealing with variable pairs that objectively were only modestly correlated. The illustration provided in Figure 2 drives this point home. Several pairs of variables correlated at levels that, in the case of data-based assessments, had yielded mean subjective estimates quite close to 0 (e.g., $r = .3$) now produced estimates of 30, 40, or more, on the subjective 100-point scale.

Variable pairs G and I merit special emphasis, since they deal with theory-based estimates of cross-situational behavioral consistency and thereby speak to an issue of particularly active theoretical interest at the present time (cf. Alker, 1972; Allport, 1966; Bem & Allen, 1974; Bem & Funder, 1978; Bem & Lord, 1979; Mischel, 1968, 1969). The message conveyed by the subjective estimates for these two relationships is clear

and dramatic. When faced with immediately available objective data presenting correlations in the range $r = .2$ to $r = .3$, subjects had provided mean estimates averaging between 0 and 10 on the 100-point rating scale. Now, in the absence of immediately available data, but dealing with bivariate domains featuring the same $r = .2$ to $r = .3$ range, subjects' implicit personality theories led them to offer covariation estimates that averaged in the 50 to 60 range. What degree of covariation in immediately available bivariate data would have been required to prompt similar estimates in the absence of "theories" or preconceptions? The answer, as we can see in Figure 2, is that only empirical correlations in the range of $r = .90$ could have prompted such estimates.

The results of our own research thus can be summarized with a pair of empirical generalizations.

1. When immediately available bivariate data are examined in the absence of any theories or preconceptions about their relatedness, the intuitive psychologist has great difficulty in detecting covariations of the magnitudes that are apt to characterize a wide range of functional relationships presented by everyday social experience. In particular, covariations in the range of magnitudes commonly dealt with by psychometricians seeking cross-situational behavioral consistencies are most likely to go undetected or be perceived as close to zero.

2. When no objective, immediately available, bivariate data can be examined, but prior theories or preconceptions can be brought to bear, the intuitive psychologist is apt to expect and predict covariations of considerable magnitude – often of far greater magnitude than are likely to have been presented by past experience or to be borne out by future experience.

Taken together, these generalizations help to sharpen the focus of the present contrast between data-based and theory-based assessments. The theories we hold apparently lead us to expect and predict stronger empirical relationships than actually exist; and many of the empirical relationships that do exist, even ones of consequential magnitude, are apt to go undetected unless we already expected to find them.

On the origin and survival of theories

Covariation assessment strategies

Beyond offering generalizations about the outcomes of various assessment tasks, it is important to begin considering the judgmental processes and strategies that might underlie such outcomes. Once again the distinction between data-based and theory-based assessment proves a critical one, and once again the Chapmans' seminal findings provide a convenient point of departure.

Data versus theories. It was clear from the Chapmans' studies that preconceptions overcame the information provided by immediately available

data. Less clear, however, was the manner in which this occurred. Did the subjects actually “see” the relationships that they reported? Or did they merely report the relationships that they expected to be present in the data, without any corresponding subjective impression? Or did the subjects “compromise” between their subjective perceptions and expectations, giving some weight to each? It is impossible to give a definitive answer on the basis of the reported results, although there is clear evidence both that subjects did give at least some weight to their immediate perceptions of the data and that their perceptions were not totally determined by their expectations. Specifically, we note the Chapmans’ report that repeated exposure to non-relationships did ultimately reduce reports of illusory correlation. However, it was also clear that the data could never totally triumph over the subjects’ preconceptions, for even extended exposure to *negative* relationships could not completely eliminate reports of illusory positive correlations.

The questions concerning intuitive strategies for covariation assessment thus come sharply into focus. How do subjects decide what degree of relationship is present in the data at hand? How do they decide what degree of relationship “ought” to be present in a representative set of bivariate observations that are addressed by their intuitive theories or preconceptions? When, and how, do preconceptions transform one’s subjective experiences of covariation? In beginning to speculate about the answers to such questions it is once again helpful to focus initially on purely data-based and purely (or at least largely) theory-based assessments, and only then return to the case when immediately available data and theory-based expectations clash.

Strategies for data-based assessment. Perhaps our most striking finding regarding data-based assessment involved the subjects’ responses to bivariate distributions offering covariations of magnitudes that are likely to be encountered in everyday experience and are relevant to informal social theories – specifically, subjects found it very difficult to detect such relationships and, on the average, they rated the relevant covariations to be quite close to zero. What implications do such results have concerning the informal covariation assessment strategies that might have been employed by our subjects?

In speculating about this issue, it is important to remember that individual raters did not generally estimate objectively weak correlations to be close to zero. Instead, they offered a fairly broad range of estimates for which only the relevant means were close to the zero point. Once again, it is difficult to discern exactly what the subjects actually “saw” and what they estimated in spite of what they saw. The strikingly smooth and consistent “psychometric” function relating mean estimates to actual correlations leaves little doubt that the subjects did make use of data characteristics that were consistently related to the objective correlation.

The amount of variability in their estimates, however, indicates that the data characteristics utilized were only very roughly related to objective measures of covariation.

Our suspicion, buttressed by the introspections of some naive raters who have undertaken the task, is that subjects' impressions of "relatedness" do not reflect any attempt to consider the total sample of relevant bivariate observations. Rather, subjects may rely upon a few special cases, probably the extreme cases of variable *X* or variable *Y*. Thus, they may simply observe the direction and extremity of the value of *Y*s associated with the extreme values of *X*, and vice versa. Factors influencing attention and memory might also play a role; thus vividness and concreteness, as well as primacy and recency of presentation, might influence which data points are relied upon.

Reliance upon a limited number of special or "test" cases in this fashion, particularly reliance upon extreme scores, would produce certain consequences that seem highly consistent with our own findings: First, manipulations of stimulus domain (i.e., numbers vs. pictures vs. audiotapes) that were designed to vary the necessity for magnitude estimation and memory prior to covariation assessment should have exerted little impact, since these additional demands were rendered trivial if the raters merely relied upon a very limited number of special cases. Second, depending upon which particular cases they happened to take note of, subjects dealing with weak relationships could be expected to estimate the relationships to be moderately positive or moderately negative rather than zero. By contrast, subjects dealing with strong relationships should have shown quite consistent and extreme estimates, since highly positive relationships permit little variability in the values of one variable that can be associated with extreme values of the other variable. The shape of our psychometric function, and the reduced variability associated with estimates of the strongest empirical relationship presented, are consistent with these postulated consequences of relying upon extreme cases, although such evidence is far from conclusive. Clearly, more definitive answers could come only from research designs that deliberately manipulated the value of extreme, or otherwise disproportionately available, cases.

Strategies for theory-based assessment. When we turn our attention to theory-based assessments, the obvious question to be posed involves the extent to which any data may have been utilized in addition to pure theory. In some cases, of course, it is probable that no data are consulted at all. Raters, for instance, may have relied solely upon semantically based or theory-based intuitions and expectations; that is, applied "representativeness" criteria (cf. Kahneman & Tversky, 1973, 4) in considering the relationships between the two variables in each pair. Few readers are likely to dispute the contention that people hold – or at least can readily generate – social theories that have such origins. Indeed, any layperson can generate new

theories or predictions about functional relationships far too rapidly for those theories to depend upon the generation and analysis of actual cases.

Nevertheless, we do not question the possibility that *some* theory-based covariation estimates may involve resort to actual evidence. Once again, however, we suspect that the layperson is far more likely to rely upon particular test cases – subject to many sources of significant biases in coding, storage, and recall – than to generate and informally analyze some overall sample of bivariate data.

Some theories or beliefs about particular relationships may thus be based at least partially on data. Others may be based on deductions from broader beliefs about the world, conventional folk wisdom, semantic associations, or persuasive communications by family, friends, or the mass media. Regardless of their origins, however, it is clear that many such beliefs are both erroneous and able to survive and perhaps even flourish in the face of evidence that would create grave doubts in any unbiased observer – certainly in any unbiased observer who owned a calculator, an introductory statistics text, and some conventional knowledge about how to use them. We shall, therefore, conclude this chapter by briefly addressing a pair of related questions about the clash between intuitive theories and everyday experiences with the empirical covariations that “test” those theories: First, what are the mechanisms by which erroneous or highly exaggerated beliefs about functional relationships can survive in the face of seemingly compelling logical or evidential challenges? Second, how can our emerging portrait of the layperson’s deficiencies be reconciled with the obvious capacity of organisms to show learning (in particular, operant and respondent conditioning) and with the obvious accuracy and adequacy of so many of our everyday beliefs and social strategies?

These questions can only be touched upon here, and we refer the interested reader to more comprehensive discussions in Chapter 9 of this volume and in other publications (e.g., Nisbett & Ross, 1980).

Mechanisms of theory perseverance

Theories about empirical relationships, like impressions about particular individuals, often show an amazing capacity to survive in the face of empirical challenges. One set of mechanisms that accounts for such perseverance has its origin in the simple fact that people characteristically act upon their beliefs; these actions may range from simple public advocacy to the investment of one’s time, energies, wealth, or reputation. Such behavioral commitment makes it highly “dissonant” (Abelson et al., 1968; Festinger, 1957, 1964) for the actor simply to update his beliefs in the light of new evidence. Indeed, an actor’s beliefs or expectations can constitute a self-fulfilling prophecy, whereby initially groundless beliefs cause the actor to behave in a manner that produces outcomes or data that

ultimately validate his beliefs (Merton, 1948; Rosenhan, 1973; Rosenthal & Jacobsen, 1968; Snyder & Swann, 1976; Snyder, Tanke, & Berscheid, 1977).

Recent research by Lepper, Ross, and their colleagues has illustrated additional mechanisms or processes that may underlie belief perseverance. These mechanisms involve the lay scientist's propensities both to assimilate newly considered items of information as a function of their consistency with prior beliefs and to go beyond the assimilation of data to the development of causal explanations capable of explaining *why* the hypothesized state of affairs would hold (see Chapter 9 of this volume and Anderson, Lepper, & Ross, 1980; Jennings, Lepper, & Ross, 1980; Lord, Ross, & Lepper, 1979; Nisbett & Ross, 1980; Ross, 1977; Ross, Lepper, & Hubbard, 1975; Ross, Lepper, Strack, & Steinmetz, 1977).

Covariation detection, conditioning, and other real-world successes

Our bleak portrait of the layperson's capacities to detect and assess covariation may seem at first blush to be incongruent with an enormous body of laboratory evidence, and an even broader base of everyday experience, illustrating classical and operant conditioning. Every rat who has ever learned to escape in a shuttle box, every child who has ever discerned the advantages of adding sugar to her breakfast cereal, every lover who has discovered that honesty is not always the best policy, every mother who has ever shown a lactation reflex in response to a child's cry, proves that organisms do recognize covariations among environmental stimuli. Does the work of the Chapmans and that of the present investigators therefore deal with some very narrow and aberrant class of covariation assessment problems at which the layperson is uniquely deficient? Nisbett and Ross (1980) have discussed this issue and contended that the answer is no. Instead, they claim, it is the conditioning phenomena that constitute the "exception," and it is the limited capacity to detect and assess covariation described throughout this chapter that illustrates the "rule."

We cannot review the relevant arguments in detail, but the gist of Nisbett and Ross's thesis is that classical and operant conditioning generally are obtained under sharply constrained circumstances involving stimulus salience and distinctiveness, optimal interstimulus and intertrial intervals, and an absence of irrelevant or distracting stimulus events. Two additional factors they cite, however, merit special emphasis. First, it is noteworthy that while conditioning can be maintained with relatively low covariation between conditioned stimulus (CS) and unconditioned stimulus (UCS) or response and reinforcement, it almost invariably is initially *obtained* with conditions of perfect covariation – that is, the CS or response is invariably followed by an UCS or reinforcement, and the latter is never presented in the absence of the former. The asymmetry between necessary conditions for acquisition of a conditioned response and the

maintenance or "perseverance" of such a response thus may illustrate, to some extent, the same asymmetry between data-based and theory-based covariation assessment that has been explored in the present chapter. That is, during maintenance the organism may continue to expect, and perhaps even to perceive, a greater covariation between CS and UCS or between response and reinforcement than objectively is justified. It may accept reinforced trials at face value while attributing non-reinforced trials to extenuating circumstances, third variable influences, or even chance.

Finally, and perhaps most important, is the mounting evidence that both operant and classical conditioning benefit from, and in some cases may even demand, a reasonably good fit between the contingencies to be learned and the prior theories or expectations that the organism brings to the laboratory. At the very least, it is clear that not all CS-UCS or response-reinforcement contingencies are equally learnable. Perhaps the most impressive demonstrations of this point have been provided by Garcia and his coworkers (e.g., Garcia, McGowan, & Greene, 1972). These investigators have reported that rats can learn, on a single trial, to avoid a novel-tasting food that is followed by gastrointestinal illness, even if the interval between eating and sickness is as much as 12 hours; by contrast, the animal that becomes ill several hours after consuming a food of familiar taste but novel shape shows no such avoidance learning. On the other hand, when immediate pain replaces delayed sickness as the UCS, an opposite pattern of results is obtained; that is, the rat readily learns to avoid novel shapes but not novel tastes when such stimuli are immediately followed by electric shock. As Nisbett and Ross (1980) summarize, the rat may be characterized as possessing two "theories," both of which are well suited to the actual contingencies of its ecology: (a) Distinctive gustatory cues, when followed by later (even much later) gastric distress, should be considered suspect; (b) Distinctive tactile or spatial cues, when followed by immediate somatic pain should be considered suspect.

The argument, in more general terms (cf. Testa, 1974) is thus that organisms – people as well as rats – are likely to see those covariations, and only those covariations, that their own history or the history of their species disposes them to see. When not guided by "theories," covariation detection becomes very difficult and is likely to occur only when the relevant correlations approach unity, and/or when the conditions for learning are optimal in terms of the very factors that have been explicated in the laboratories of a long and distinguished line of Hullians and Skinnerians.

Sometimes, of course, everyday circumstances *are* optimal for learning. Thus, in our everyday experience we learn what countless switches, levers, buttons, and other manipulanda do, and what a bewildering variety of signs, symbols, and signals mean, because the relevant covariations are so close to perfect. Equally important, perhaps, is the fact that the layperson, like the formal scientist, often can "test" new hypotheses that

he has come to entertain. Sample data can deliberately be generated that are far better suited for the inferential task at hand than the samples offered by accidental experience and haphazard recollection.

Indeed, our mastery of our environment has increasingly depended upon our capacity to substitute relatively formal tools of inference for informal ones. Our success reflects the legacy of generations of ordinary men and women who have carefully noted and recorded their findings and, more recently, that of countless scientists trained in the arts of formal experimentation and statistical analysis.